

Mutual Fund Analytics

End-to-End Data Engineering, Performance Analytics & Interactive Dashboard

Final Project Report

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August 2026

Project scope	Coverage
Mutual fund analytics platform	40 schemes, 10 datasets, 4.5+ years of NAV history
Core stack	Python, Pandas, NumPy, SciPy, SQL/SQLite, Tableau
Outputs	ETL pipeline, analytical notebooks, risk analytics, 4-page dashboard

Educational project. Mutual fund investments are subject to market risks. This report is for analytical and skill-development purposes and is not financial advice.

1. Executive Summary

A concise view of what was built and what the analysis shows.

This project builds an end-to-end mutual fund analytics workflow for Bluestock Fintech. The pipeline starts with public mutual fund and market data, cleans and validates the datasets, stores structured analytical data, computes fund and investor metrics, and presents the results through an interactive four-page Tableau dashboard.

The analysis covers 40 mutual fund schemes, 10 provided datasets, roughly 46,000 NAV observations, and about 32,000 investor transaction records. The project also incorporates benchmark index data and portfolio holdings. The analytical workflow was designed around eight objectives: ETL, normalized data modeling, EDA, performance and risk metrics, dashboarding, investor analysis, benchmark comparison, and final documentation.

At the industry level, the project uses December 2025 reference points of approximately ₹81 lakh crore industry AUM, 1,908 schemes, 26.12 crore folios and ₹31,002 crore monthly SIP inflow. The Tableau Page 1 intentionally shows ₹62.74 lakh crore as the AUM of the top 10 AMCs represented in the source dataset, alongside the wider industry benchmark of about ₹81 lakh crore.

The fund-performance workflow evaluates return, volatility, Sharpe, Sortino, Alpha, Beta and maximum drawdown. A composite fund score uses 3-year CAGR, Sharpe, Alpha, inverse expense ratio and inverse drawdown ranks. This keeps fund selection from becoming a single-metric ranking exercise.

The main limitation is data authenticity. AMFI, benchmark and fund metadata are based on public sources, while investor transactions are synthetic and some NAV history is anchored to real observations and simulated forward. The results should therefore be treated as an educational analytical demonstration rather than an investment recommendation.

2. Problem Statement & Objectives

Mutual fund analysis combines multiple data types: scheme metadata, NAV histories, AUM, SIP activity, investor transactions and benchmark prices. Without a common pipeline, these datasets are difficult to validate, compare and present consistently. The project addresses that problem by creating one repeatable analytical workflow.

Objective	Expected outcome
O1 — ETL pipeline	Automated ingestion, cleaning and transformation of source datasets.
O2 — SQL data model	Normalized relational structure with dimensions and fact tables.
O3 — EDA	Trend, distribution, correlation and industry/investor insights.
O4 — Performance & risk	Return, Sharpe, Sortino, Alpha, Beta, drawdown and scorecard.
O5 — BI dashboard	Four-page interactive Tableau dashboard.
O6 — Investor analysis	Transaction, geography and demographic analysis.
O7 — Benchmark comparison	Fund versus NIFTY benchmark analysis and tracking error.
O8 — Documentation	Final report, presentation and reproducible repository.

3. Data Sources & Dataset Inventory

The project brief identifies AMFI India, mfapi.in, NSE India, BSE India and AMFI Monthly Notes as primary public sources. It also includes pre-packaged CSV datasets supplied for the capstone. No proprietary or confidential data is required.

Dataset	Approx. size	Purpose
01_fund_master.csv	40 rows	Scheme metadata, AMFI code, fund house, category, expense ratio and risk grade.
02_nav_history.csv	≈46K rows	Daily NAV history for 40 schemes, Jan 2022–May 2026.
03_aum_by_fund_house.csv	≈90 rows	Quarterly AUM for the top 10 fund houses.
04_monthly_sip_inflows.csv	48 rows	Monthly SIP inflow, active accounts and registrations.
05_category_inflows.csv	≈144 rows	Monthly/category net inflows for FY25 analysis.
06_industry_folio_count.csv	21 rows	Industry folio milestones.
07_scheme_performance.csv	40 rows	Return and risk metrics derived from NAV history.
08_investor_transactions.csv	≈32K rows	SIP, Lumpsum and Redemption transactions for 5,000 investors.
09_portfolio_holdings.csv	≈320 rows	Equity holdings, weights and sectors.
10_benchmark_indices.csv	≈8K rows	NIFTY and other benchmark closing values.

Data quality note: the ingestion work validated 40 AMFI codes across the fund master and NAV history, found no duplicate rows, and reported missing values only in the yoy_growth_pct field of the monthly SIP dataset. Date fields were standardized during preprocessing.

4. System Architecture & ETL Design

The project follows an Extract → Transform → Load → Analyse → Visualise workflow. This separates source acquisition from cleaning, storage, analytics and dashboard presentation, making it easier to validate each stage independently.

Layer	Implementation	Key operations
Extract	AMFI, mfapi.in, NSE/BSE, supplied CSVs	Load source files, fetch NAV API data, capture benchmark and industry data.
Transform	Python + Pandas	Clean types, normalize names, handle missing NAV days, validate AMFI codes, derive returns and aggregates.
Load	SQLite / structured CSVs	Store dimensions and facts around funds, dates, NAV, transactions and performance.
Analyse	Jupyter + NumPy + SciPy	CAGR, Sharpe, Sortino, Alpha, Beta, drawdown, VaR/CVaR, cohort and concentration analysis.
Visualise	Tableau	Four dashboards with filters, scorecard, benchmark comparison and market/industry views.

Relational design

The project specification uses a star-schema approach with dimensions such as fund and date, and facts such as NAV, transactions, performance, AUM, portfolio holdings and industry SIP data. AMFI code and date are the primary linking concepts for the analytical model.

5. Data Preparation, Validation & ETL

The ETL stage was designed to prevent downstream dashboard errors. Source tables were inspected for shape, data types, missing values, duplicates and key consistency before analytical transformations were applied.

- Fund master and NAV history contain 40 matching AMFI codes. This validates the core fund-to-NAV relationship used for performance analysis.
- The ingestion audit reported zero duplicate rows across the loaded datasets.
- Missing values were identified in the monthly SIP yoy_growth_pct field. The field is derived and does not invalidate the underlying SIP inflow series.
- Date columns initially represented as strings were converted during preprocessing so that time-series grouping and joins could be performed consistently.
- NAV observations require calendar awareness because weekends and market holidays do not have trading observations. The project methodology accounts for these gaps during return calculations.

The result is a cleaned analytical layer that can feed both notebook analysis and Tableau without rebuilding the same transformations separately.

6. EDA Findings: Industry & Market Activity

The industry analysis shows a clear expansion in AUM, SIP participation and folio count over the project period. The dashboard uses quarterly AMC AUM and monthly SIP/folio data to provide complementary views of industry scale.

Indicator	Observed reference point	Interpretation
Industry AUM	≈₹81 lakh crore, Dec 2025	Wider industry benchmark used to contextualize the top-10 AMC view.
Top-10 AMC AUM	₹62.74 lakh crore in the dashboard	Represents the AMCs covered by the source dataset, not the entire industry.
Monthly SIP inflow	₹31,002 crore, Dec 2025	Shows strong systematic-investment activity at the end of the analysis window.
Total folios	26.12 crore, Dec 2025	Indicates broad retail participation in the mutual fund industry.
Largest AMC in dataset	SBI Mutual Fund, ₹12.50 lakh crore	Highlights the concentration of AUM among leading fund houses.

- SIP inflow rises substantially over the 2022–2025 window, with Dec 2025 reaching ₹31,002 crore.
- The AUM-by-AMC view shows a pronounced scale difference between the largest and smaller AMCs in the selected top-10 set.
- The analysis should distinguish industry-wide benchmarks from dataset-covered aggregates. This distinction is explicitly shown on the dashboard to avoid presenting ₹62.74 lakh crore as total industry AUM.

7. EDA Findings: Investor & Category Behaviour

The investor dataset contains SIP, Lumpsum and Redemption transactions for 5,000 investors across 12 states, with city tier and age-group attributes. This allows the dashboard to combine financial activity with basic demographic and geographic context.

The dashboard shows transaction amounts by state, the transaction-type split, average SIP amount by age group and monthly transaction volume. The observed average SIP amounts are close across age groups, with the 56+ group highest in the dashboard at approximately ₹11,575 and the 46–55 group at approximately ₹11,137.

Category analysis adds a second lens. The FY25 top-five net inflow categories shown on the dashboard are Liquid (₹451,275 Cr), Sectoral/Thematic (₹103,829 Cr), Flexi Cap (₹63,989 Cr), Large & Mid Cap (₹57,752 Cr) and Short Duration (₹55,530 Cr).

- The category heatmap is useful for spotting persistence or changes in monthly flow intensity rather than relying on a single FY25 total.
- State-level transaction totals reveal where recorded investor activity is concentrated, but the synthetic nature of the dataset means these patterns should not be treated as actual customer-market estimates.
- Transaction-type comparisons are best used as behavioural demonstrations within the capstone rather than forecasts of actual Indian investor allocation.

8. Fund Performance & Risk Methodology

Performance analysis is built from daily NAV observations. Daily returns are calculated from consecutive NAV values and then aggregated into multi-period returns and risk measures.

Metric	Method	Use
1Y / 3Y / 5Y return	Period return / CAGR from NAV history	Measures absolute and sustained performance.
Sharpe ratio	$(\text{Return} - \text{risk-free rate}) / \text{volatility}$; 6.5% proxy	Risk-adjusted return using total volatility.
Sortino ratio	Excess return divided by downside volatility	Focuses risk measurement on negative returns.
Alpha	OLS intercept against NIFTY 100, annualized	Measures benchmark-relative excess return.
Beta	OLS slope against benchmark returns	Measures sensitivity to market movements.
Maximum drawdown	Minimum NAV / running peak – 1	Captures worst peak-to-trough decline.
Tracking error	Std. deviation of fund vs benchmark return differences	Measures consistency of benchmark-relative performance.

9. Multi-Factor Scorecard & Advanced Analytics

The fund scorecard combines several dimensions instead of ranking funds by return alone. The composite score uses 30% 3-year CAGR rank, 25% Sharpe rank, 20% Alpha rank, 15% inverse expense-ratio rank and 10% inverse maximum-drawdown rank.

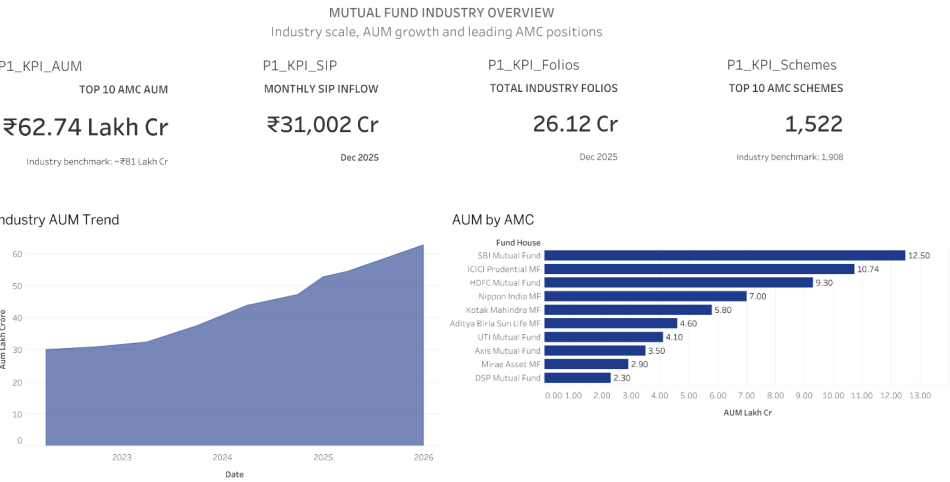
The selected benchmark-comparison set includes Mirae Asset Large Cap Fund, ICICI Pru Midcap Fund, HDFC Mid-Cap Opportunities Fund, Kotak Flexicap Fund and ICICI Pru Bluechip Fund. These funds were rebased against NIFTY 50 and NIFTY 100 for comparison.

The advanced analytics stage also covers historical VaR and CVaR at 95%, rolling 90-day Sharpe, investor cohort analysis, SIP continuation analysis, a simple risk-grade-based recommendation function and sector concentration using the Herfindahl-Hirschman Index.

Advanced analysis	Purpose
VaR / CVaR	Estimate historical tail-loss thresholds and average losses beyond the VaR threshold.
Rolling Sharpe	Show how risk-adjusted performance changes over time rather than using one static ratio.
Investor cohorts	Compare investors by first transaction year, average SIP amount and investment behaviour.
SIP continuity	Flag long transaction gaps above 35 days for investors with sufficient SIP history.
Sector HHI	Quantify portfolio concentration, where higher HHI indicates more concentrated sector exposure.

10. Dashboard Results — Page 1: Industry Overview

Page 1 combines four executive KPIs with an industry AUM trend and a top-10 AMC comparison. It makes the dataset-vs-industry distinction explicit. ₹31,002 Cr monthly SIP inflow and 26.12 Cr folios are December 2025 reference points. The ₹62.74 lakh crore AUM figure is the top-10 AMC aggregate available in the source data, while approximately ₹81 lakh crore is shown as the wider industry benchmark.



Methodology: AUM and scheme metrics represent the Top 10 AMCs available in the source dataset. Industry benchmarks are shown for comparison where applicable.

Source: completed Tableau dashboard export.

11. Dashboard Results — Page 2: Fund Performance

Page 2 brings the fund-selection workflow into one view: return-versus-risk scatter, sortable scorecard and normalized NAV versus benchmark. Fund House, Category and Plan slicers allow the user to narrow the analysis. The NAV-versus-benchmark view uses a normalized base of 100.

FUND PERFORMANCE

Risk, return and fund-level performance analytics

Fund House

All

Category

All

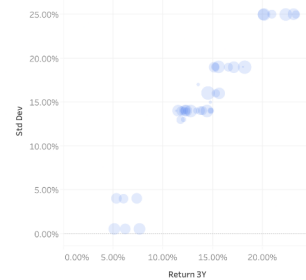
Plan

All

Scheme Name

ABSL Frontline Equity Fund - Regular - Growth

Performance Scatter



Fund Performance Scorecard

Scheme Na...	Fund House	Category	Plan	Return 1Y	Return 3Y	Return 5Y	Std Dev	Sharpe Rat...	AUM	Alpha	Bet
SBI Small C...	SBI Mutual ...	Small Cap	Regular	24.56%	23.39%	20.67%	25.00%	0.94	19,259	1.23	
SBI Small C...	SBI Mutual ...	Small Cap	Direct	20.59%	23.14%	21.82%	25.00%	0.93	36,061	1.13	
ABSL Small ...	Aditya Birla...	Small Cap	Regular	24.93%	22.38%	23.60%	25.00%	0.90	41,613	1.84	
Axis Small C...	Axis Mutual...	Small Cap	Regular	21.97%	20.98%	22.62%	25.00%	0.84	21,545	0.51	
Nippon Indi...	Nippon Indi...	Small Cap	Regular	21.30%	20.15%	21.88%	25.00%	0.81	43,630	0.80	
DSP Small C...	DSP Mutual...	Small Cap	Regular	20.20%	20.08%	20.61%	25.00%	0.80	35,124	0.69	
Kotak Emer...	Kotak Mahi...	Mid Cap	Regular	17.12%	18.23%	17.75%	19.00%	0.96	47,469	1.91	
ICICI Pru M...	ICICI Prud...	Mid Cap	Regular	14.02%	18.08%	17.55%	19.00%	0.95	979	0.89	
DSP Midcap...	DSP Mutual...	Mid Cap	Regular	14.12%	17.16%	19.00%	19.00%	0.90	37,835	1.02	
HDFC Mid-C...	HDFC Mutu...	Mid Cap	Regular	15.43%	16.58%	17.69%	19.00%	0.87	23,185	0.95	
Kotak Flexic...	Kotak Mahi...	Flexi Cap	Regular	15.74%	15.65%	13.50%	16.00%	0.98	35,012	1.85	
UTI Mid Cap...	UTI Mutual ...	Mid Cap	Regular	16.30%	15.61%	15.86%	19.00%	0.82	41,728	1.12	
UTI Flexi Ca...	UTI Mutual ...	Flexi Cap	Regular	17.43%	15.34%	15.78%	16.00%	0.96	17,912	1.79	
HDFC Mid-C...	HDFC Mutu...	Mid Cap	Direct	19.98%	15.29%	15.85%	19.00%	0.80	18,792	0.90	
Axis Midcap...	Axis Mutual...	Mid Cap	Regular	14.88%	15.19%	18.94%	19.00%	0.80	28,995	1.42	
HDFC Top L...	HDFC Mutu...	Large Cap	Regular	10.94%	14.84%	11.32%	14.00%	1.06	6,434	0.78	
Mirae Asse...	Mirae Asse...	Large Cap	Regular	15.12%	14.81%	12.68%	14.00%	1.06	11,361	1.62	
ICICI Pru Va...	ICICI Prud...	Value	Regular	16.67%	14.76%	12.60%	15.00%	0.98	2,571	0.55	
Mirae Asse...	Mirae Asse...	Large & Mid	Regular	14.91%	14.56%	15.68%	16.00%	0.91	49,046	1.70	

NAV VS BENCHMARK (NORMALIZED BASE = 100)

NAV vs Benchmark (Normalized Base = 100)



Source: completed Tableau dashboard export.

12. Dashboard Results — Page 3: Investor Analytics

Page 3 focuses on recorded investor behaviour. It combines transaction amount by state, transaction type split, average SIP amount by age group and monthly transaction volume, with State, Age Group and City Tier filters. Because the investor data is synthetic, these charts demonstrate the analytical workflow rather than verified customer behaviour.

INVESTOR ANALYTICS

Transaction behaviour, investor segments and monthly activity

State

All

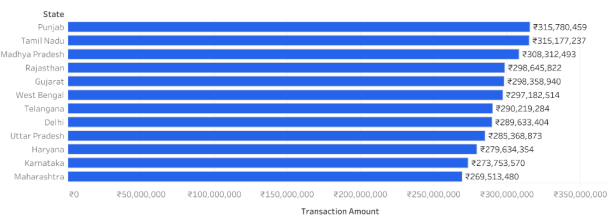
Age Group

All

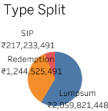
City Tier

All

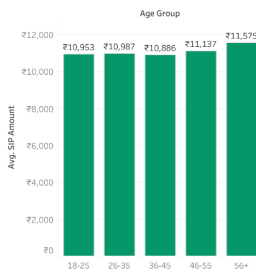
Transaction Amount by State



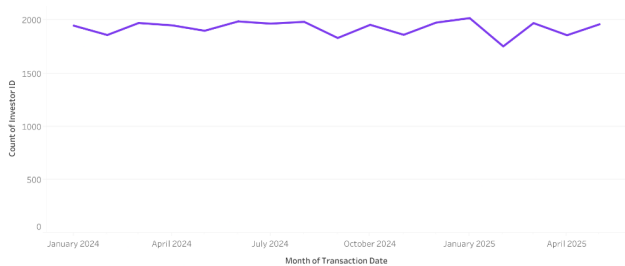
Transaction Type Split



Age Group vs Average SIP Amount



Monthly Transaction Volume



Source: completed Tableau dashboard export.

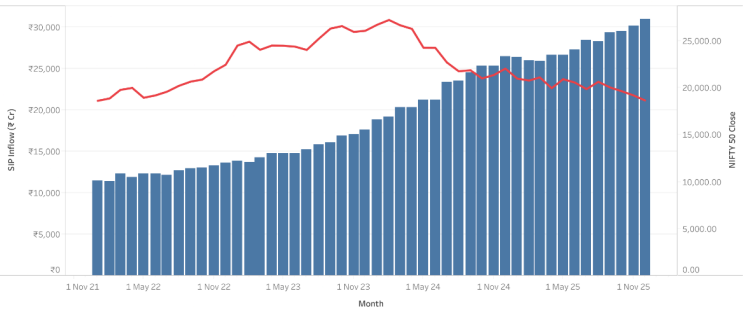
13. Dashboard Results — Page 4: SIP & Market Trends

Page 4 connects systematic investment activity with market context. The dual-axis view overlays monthly SIP inflow bars with the NIFTY 50 line, while the lower section shows category-level net inflow intensity. The FY25 top-five category ranking is led by Liquid, followed by Sectoral/Thematic, Flexi Cap, Large & Mid Cap and Short Duration.

SIP & MARKET TRENDS

Systematic Investment Plan inflows, NIFTY 50 correlation and category-wise net flows

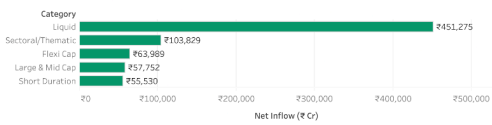
SIP Inflow vs NIFTY 50 Monthly Trend (2022 - 2025)



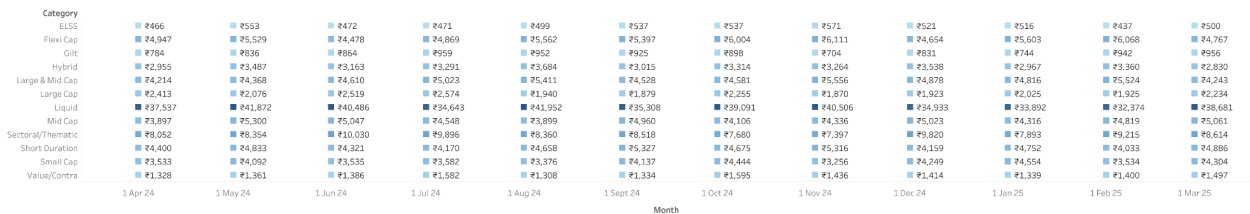
SIP Accounts Growth (YoY)

13.7%

Top 5 Categories by FY25 Net Inflow (₹ Cr)



Monthly Category Net Inflow Heatmap (₹ Cr)



Source: completed Tableau dashboard export.

14. Limitations, Data Authenticity & Responsible Use

The project is designed as a capstone analytics platform, so the distinction between real, simulated and synthetic information is important.

Area	Limitation	Implication
Investor transactions	Synthetic transactions for 5,000 investors.	Investor behaviour findings are illustrative, not market estimates.
NAV history	Anchored to real historical values with simulated forward values in parts of the dataset.	Long-horizon performance results should not be treated as verified live performance.
Industry AUM	Dashboard top-10 AMC aggregate is ₹62.74 lakh crore, while the brief cites ₹81 lakh crore industry AUM.	The two figures must not be presented as the same measure.
Benchmarking	Benchmark comparisons depend on selected date ranges and benchmark series.	Results are context-specific rather than universal rankings.
Investment use	No portfolio suitability, tax or personal risk assessment is performed.	The dashboard is an analytical tool, not an investment advisory system.

Responsible-use statement

All fund-selection observations should be interpreted as analytical outputs from the supplied dataset. Mutual fund investments are subject to market risks, and historical or simulated performance does not guarantee future results.

15. Recommendations & Conclusion

Recommendations

- Keep the dashboard centered on a small number of decision metrics: return, Sharpe/Sortino, Alpha, drawdown and benchmark-relative performance.
- Maintain explicit dataset labels for industry-wide benchmarks versus top-10 AMC aggregates so that stakeholders do not compare unlike measures.
- For a production deployment, replace synthetic investor transactions and simulated forward NAV values with governed, refreshed data sources.
- Automate the ETL and validation checks so that new monthly SIP, AUM, folio and benchmark data can be incorporated without manual rework.
- Add stronger dashboard drill-through and published access controls if the project moves from capstone evaluation to operational use.

Conclusion

The project delivers the requested analytical chain from raw mutual fund data through ETL, SQL modeling, EDA, performance and risk analytics, investor analysis and an executive Tableau dashboard. The strongest outcome is the separation of analytical stages: data quality is checked before metrics are computed, performance is evaluated using multiple risk-adjusted measures, and dashboard views are tied to defined calculations.

The resulting platform is suitable as a portfolio-quality demonstration of data engineering and analytics skills, while its limitations are clearly documented so that the dashboard is not mistaken for a production investment-advisory system.

Primary project sources: AMFI India, mfapi.in, NSE/BSE public market data, AMFI Monthly Notes and the datasets supplied with the Bluestock Fintech capstone. Technical methodology follows the official project brief and the completed analytical outputs.